Part 3 — Getting the product out: toxicity, in situ removal, two-phase modelling and immobilisation (weeks 45 to 48)

These four weeks have no textbook. Not in the sense that the books are old or badly pitched, which is the complaint about phase 2, but in the sense that no book covers them: in situ product removal lives in a few hundred papers, two decades of the Daugulis group's polymer work, and scattered chapters of bioprocess texts that treat it as an aside. The modelling in week 47 exists mostly as supplementary material. So this part is the primary text rather than a spine, and it is written accordingly — the derivations are here in full, the models are complete enough to run, and where I give a number I also give the parameters that produced it.

The good news is that this is your own subject. Partition coefficients, isotherms, intraparticle diffusion, Thiele moduli, effectiveness factors, washout and dilution rate are chemical engineering, and the biologists in the field are mostly learning them from scratch. If you want one block of the curriculum where a chemical engineer arrives with an unfair advantage, it is this one. The things to be careful about are not the transport calculations but the biology wrapped around them: that the cells are in one phase and not the other, that a solvent can be eaten, and that a bead full of cells runs out of oxygen long before it runs out of sugar.

A warning about vocabulary, because the literature is loose with it and the confusion is the commonest error in this area. In situ product removal is the *goal*. A two-phase partitioning bioreactor is one *reactor* that achieves it. Liquid–liquid means a solvent; solid–liquid means polymer beads or resin. Immobilising the cells is a different thing entirely, aimed at a different problem, and it is week 48. Keep those separate and half the papers become easier to read.

Ch.	Week	Chapter
45	45	Product toxicity, export and loss: deciding what actually limits the titer
46	46	In situ product removal: choosing a strategy, and sizing a sorbent
47	47	Modelling a two-phase fermentation, and fitting it to data
48	48	Cell immobilisation: diffusion, dead cores and continuous operation

Product toxicity, export and loss: deciding what actually limits the titer

Three different things stop a titer, they look identical on an HPLC trace, and they call for three different interventions. Before choosing a lever, find out which one you have — it takes a week and saves a year.

The three limits, and the experiments that separate them

Toxicity means the product inhibits the cells: growth slows, specific productivity falls, and the culture plateaus with substrate left in the broth. **Export** means the product is made but cannot leave: intracellular concentration rises, flux falls as the pathway is inhibited or the exporter saturates, and the broth titer plateaus at a value that is low relative to what the cells contain. **Loss** means the product leaves the vessel or decomposes: production continues at full rate and the titer plateaus anyway, because what you are measuring is the balance between making and losing.

All three produce the same curve. Three experiments distinguish them, and none takes more than a few days.

- **Spike the product into the non-producing host.** Grow the parent strain with product added at the concentrations you are reaching, and measure growth rate and, if you can, the specific productivity of some other function. If growth falls at the titer you plateau at, toxicity is real; if it does not, toxicity is not your limit, whatever the review articles say about this compound class.
- **Measure intracellular and extracellular product separately.** Quench, separate cells from broth quickly, extract both. A high intracellular concentration with a modest broth titer is an export limit. The measurement is fiddly — the product partitions during the separation — but the answer is unambiguous enough to be worth it, and it is the only direct evidence for an export limitation.
- **Compare a closed vessel with a vented one.** For anything volatile, run the same fermentation with the off-gas condensed or trapped and measure what the trap caught. The difference between the broth titer and the total made is the loss, and the experiment also tells you how much of your "poor strain" was actually a collection problem.

Write down, before you run them, what each result would mean and what you would do next in each case. That discipline — the decision tree first, the experiment second — is the difference between diagnosis and data collection, and it is the habit this week exists to install.

Why hydrophobic products are toxic, and what log P is for

A hydrophobic product does not poison a specific target. It partitions into the lipid bilayer in proportion to its hydrophobicity, swells and fluidises the membrane, and the membrane stops holding a proton gradient. That mechanism explains the pattern you will see in the literature: toxicity tracks the compound's hydrophobicity rather than its chemistry, with the worst offenders being small amphiphilic molecules — short-chain alcohols, terpene alcohols, phenolics — that are soluble enough in water to reach the cells and hydrophobic enough to accumulate in the membrane once there. A very hydrophobic compound, such as a long-chain hydrocarbon or a carotenoid, is often less toxic simply because its aqueous concentration can never get high.

Hence the two numbers of this week, which are routinely conflated and are not the same thing. **The solvent's log P** — its own octanol–water partition coefficient — is a rule-of-thumb guide to whether the cells tolerate the solvent: below about 2 solvents tend to be toxic, above about 4 they tend to be tolerated, with a grey zone between and genuine exceptions. **The product's partition coefficient K** between the solvent and water is what decides how much product the second phase captures. You need a solvent that scores well on both, and the two are only loosely related, so a solvent chosen for one can fail on the other. Write both numbers down for every candidate solvent before you order any of them.

Efflux pumps, and the honest record

If the limit is export or toxicity, a pump is the obvious genetic lever. Three families matter.

Family	Architecture	Energy	Notes for a production strain
RND	Tripartite: inner-membrane pump, periplasmic adaptor, outer-membrane channel (AcrAB–TolC)	Proton motive force	Gram-negatives only; spans both membranes, so it exports to the medium in one step. All three components must be present and balanced
MFS	Single inner-membrane protein	Proton motive	Widely distributed, simple to express; in Gram-negatives it only reaches the periplasm unless coupled to an outer-membrane partner

		force	
ABC	Transmembrane domains plus nucleotide-binding domains	ATP	Works in eukaryotes; in yeast the pleiotropic drug resistance pumps are the usual candidates. Direct ATP cost, so it shows in the energy balance

The result to know is Dunlop and colleagues' 2011 screen: 43 pumps tested against seven biofuel compounds in *E. coli*. Pumps that improved tolerance were found for five of the seven, and none helped for n-butanol or isopentanol. Read that as the field's honest base rate. Pump engineering is a screening problem, not a design problem: you cannot predict from a compound's structure which pump will move it, so you screen a panel, and sometimes the answer is that no pump in the panel works.

Pumps also cost. Expression burden is the obvious one — these are large membrane proteins, and over-expressing a membrane protein competes for the insertion machinery as well as the proteome. Substrate range is the subtler one: a pump that exports your product may also export a nutrient or an intermediate, and a tripartite pump needs its adaptor and channel at matched levels or the assembled complex is a leak in the outer membrane rather than a pump. Measure whether it paid the way you would measure any intervention: growth rate with and without, product in and out of the cell, and the titer against the same strain without the pump.

Membrane composition is the other lever, and it is worth knowing which parts are established. Shifting the saturated-to-unsaturated fatty acid ratio and cis–trans isomerisation of unsaturated chains are real, documented adaptations to solvents in several bacteria and have been used deliberately. Hopanoid and sterol engineering is research-stage. Treat the first as an option and the second as a literature you can cite rather than a tool you can use.

The second phase, arithmetically

Now the calculation that makes this week quantitative. Put a volume fraction φ of solvent into the vessel, with the product partitioning at $K = C_{org}/C_{aq}$ at equilibrium. The cells live in the aqueous phase and see only C_{aq}, so if toxicity caps the aqueous concentration at P*, the total product per litre of aqueous phase is

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P_total/L_aqueous = P* · R,        R = 1 + K·φ/(1 − φ)
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and per litre of whole broth — the number a plant cares about — it is P*·[(1 − φ) + Kφ]. Those two differ by a factor of (1 − φ) and mixing them up is the commonest arithmetic error in this subject; say per litre of what, every time.

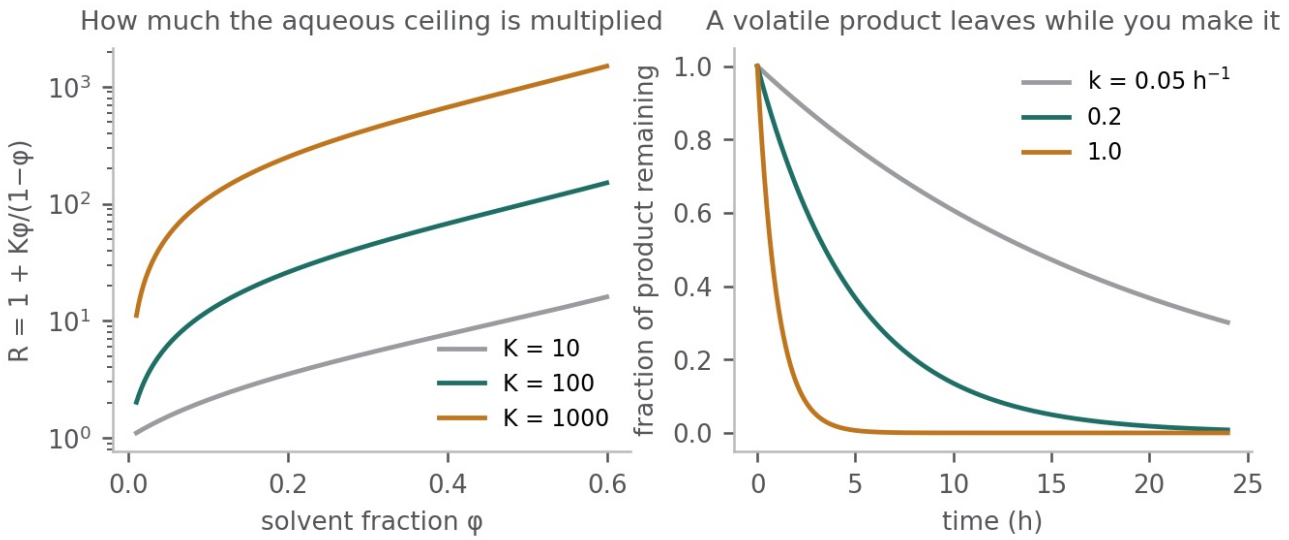


Figure 45.1 Left: the factor by which a second phase multiplies the aqueous ceiling, R = 1 + Kφ/(1 − φ), for three partition coefficients. Right: the fraction of a volatile product remaining in the broth under first-order loss at three rate constants. Both are arithmetic; the loss constants are illustrative, and measuring your own is a one-day experiment.

The numbers are striking and that is the trap. At K = 100 and φ = 0.2, R = 26: an aqueous ceiling of 2 g/L becomes 52 g/L per litre of aqueous phase, or 41.6 g/L of whole broth. At K = 1,000 and φ = 0.3, R = 430. Three things stop this being free money, and you should check all three before quoting such a number.

- **Solubility in the solvent.** R assumes the organic phase can hold $K \cdot C_{aq}$. At K = 1,000 and an aqueous concentration of 1 g/L, the organic phase must hold 1 kg/L, which no solvent can. The capture is capped by the product's solubility in the solvent, and above that the partition coefficient is irrelevant.

- **Displaced working volume.** The cells are in the aqueous phase, so a 30% solvent fraction removes 30% of your productive volume. Per litre of whole broth, output rises with φ at first and then falls; chapter 47 computes where the optimum sits for one specific model, and it was at a lower φ than the partition arithmetic alone suggests.
- **Everything else the solvent does.** Oxygen transfer through an organic layer, emulsions that will not separate, foaming, solvent loss to the off-gas, and the recovery step that has to get your product back out of the solvent. These are the subject of chapter 46 and week 49.

Loss, and the titer you measure from a culture that is producing well

For a volatile product, treat loss as first order in the broth concentration with a rate constant k that bundles stripping by the air flow, headspace loss and any degradation. Then a culture producing at a constant volumetric rate q reaches a steady titer of q/k, not a rising one. At q = 0.5 g/L/h, a loss constant of 0.05 h^{-1} gives 10 g/L, 0.2 h^{-1} gives 2.5 g/L, and 1 h^{-1} gives 0.5 g/L. The strain is identical in all three cases.

That is the single most useful equation in this chapter, because it says a plateau is not evidence that production stopped. The fraction remaining after a hold with no production is e^{-kt}: at k = 0.2 h^{-1}, 82% survives an hour, 30% survives six hours and essentially nothing survives overnight. So a sample held on the bench, or a fermentation sampled at the end of a shift, can misreport a good strain as a poor one — which is why the closed-vessel comparison is on the list of three experiments above, and why aeration rate belongs in the methods section of any volatile-product paper.

What the solvent costs

Solvent inventory is φ·V, so a 100 m³ vessel at φ = 0.2 holds 20 m³ of solvent. At a dodecane-like price of a few dollars a litre that is tens of thousands of dollars of inventory, which is acceptable as a one-off and ruinous if it is consumed. The cost per kilogram of product therefore depends almost entirely on how many cycles the solvent survives and how much is lost per cycle: inventory price divided by (kilograms of product per cycle × number of cycles), plus the make-up for losses. Work it both ways — perfect recycling and 5% loss per batch — and you will usually find that loss, not purchase, dominates. This is the same arithmetic as the resin cycles in chapter 46 and it is worth doing before the lab work rather than after.

Capture factor	`R = 1 + K·φ/(1 − φ), per litre of aqueous phase`
Per litre of broth	`P_total = P*·[(1 − φ) + K·φ]`
Volatile steady state	`P_ss = q/k; a plateau is loss, not a stopped pathway`
Hold-time loss	`P(t) = P₀·e^(−kt)`
Two different numbers	`solvent log P: will the cells live. product K: will it be captured`
Solvent cost	`inventory φ·V; cost/kg ≈ price/(kg per cycle × cycles) + make-up`

Classify four products by their main limit, and say what each calls for: a limonene-like monoterpene, a volatile sesquiterpene, a short-chain organic acid, and a flavonoid.

Monoterpene. Toxicity, and severe: it is hydrophobic enough to load the membrane and soluble enough to reach it, which is the worst combination, and aqueous ceilings of tens to a few hundred milligrams per litre are usual. Calls for a second phase from the start — an overlay is standard — plus tolerance work; a pump is worth screening but the base rate in the Dunlop data is sobering.

Volatile sesquiterpene. Loss first, toxicity second. It is less water-soluble than the monoterpene, so the cells see less of it, but it leaves with the off-gas. Calls for an overlay, which conveniently solves both — this is exactly the two-phase amorphadiene work — and for the closed-vessel experiment before concluding anything about the strain.

Short-chain organic acid. Export and toxicity, and a solvent is the wrong answer: it is charged at fermentation pH, so it does not partition into an organic phase at all. Calls for the week 14 analysis — an exporter is mandatory, diffusion will not do it — plus pH control, and if removal is wanted, ion exchange or electrodialysis rather than a solvent.

Flavonoid. Usually neither toxicity nor loss: the limits are precursor supply and, often, degradation or oxidation in the broth. Calls for checking stability in medium before anything else — spike it, incubate it without cells, and measure the decay — because if the half-life in the broth is hours, an adsorbent resin earns its place and no amount of pathway work will.

Three things people get wrong

- ‘The titer plateaued, so the product is toxic.’ Plateau is equally consistent with loss at q/k, or with an export limit. Three experiments distinguish them.

- ‘High log P means the solvent captures the product well.’ log P describes the solvent's tolerance by the cells; K describes capture. Different numbers.
- ‘Add a pump.’ The published screen found nothing useful for two of seven biofuels. A pump is a hypothesis to screen, not a fix to apply.

Self-check — answer on paper first

1. Your culture plateaus at 1.2 g/L with 40 g/L glucose left. Give the three candidate explanations and the cheapest experiment for each.

Toxicity — spike 1.2 g/L of product into the parent strain and measure growth rate. Export limitation — quench and measure intracellular against extracellular product. Loss — repeat with the off-gas trapped, or simply measure the product's half-life in medium at process temperature with no cells. The residual glucose is itself informative: it says the cells stopped consuming, which points toward toxicity or a stalled pathway rather than pure loss, since a culture losing product to the headspace would keep eating.

2. A solvent has log P = 1.8 and your product partitions into it at K = 400. Would you use it?

Not directly. A log P below about 2 puts it in the range where solvents are usually toxic, and the excellent partition coefficient does not help if the cells die — and worse, a solvent in that range is often water-miscible enough to reach the cells rather than staying as a separate phase. Options: find a more hydrophobic solvent with an acceptable K, use the toxic solvent in an external loop where the cells never contact it, or use a solid sorbent instead, which sidesteps solvent toxicity entirely and is chapter 46's main subject.

3. Derive, in one line, why the total product per litre of whole broth can fall as you add more solvent even though R rises.

Because the cells occupy only the aqueous fraction: output per litre of broth is proportional to $(1 - \varphi)$ times what the aqueous phase can hold, which is $P^{*} \cdot R$. R rises roughly as $K\varphi$ while the productive volume falls as $(1 - \varphi)$, so the product of the two has a maximum at an intermediate φ, and beyond it the displaced working volume costs more than the extra capture earns.

What to open a book for. There is no book. Read Dunlop et al., Mol. Syst. Biol. (2011) for the pump screen and its negative results, and the two amorphadiene papers — Newman et al., Biotechnol. Bioeng. (2006) and Tsuruta et al., PLoS ONE (2009) — for a two-phase process reported with real numbers. Doran's chapters on mass transfer are worth having open for the transport side, and week 14 of this module is the other half of the export argument.

In situ product removal: choosing a strategy, and sizing a sorbent

The product's own properties choose the removal method. Partition coefficient, volatility, charge, solubility and heat sensitivity decide it — not what the last paper you read happened to use.

Four reasons to remove a product as it is made

They are different reasons, they are often confused, and the one that applies determines what counts as success.

- **It inhibits the cells.** Removal raises the achievable total by keeping the aqueous concentration below the inhibition threshold. Success is measured as total product, and the control that matters is the same strain without removal.
- **It is volatile and lost.** Removal is capture: you were making it all along and now you keep it. Success is measured as recovery, and a large apparent "improvement" here is really a correction to your previous measurement.
- **It degrades in the broth.** Removal shortens the time the product spends in a hostile aqueous environment. Success is measured as product surviving to harvest, and the relevant pre-experiment is the half-life in medium without cells.
- **It inhibits its own pathway, or holds back an equilibrium.** Removal lowers Q in the week 14 sense and pulls the reaction forward. Success shows as a higher rate, not merely a higher total, and the diagnostic is whether the intracellular intermediate pools fall.

Note that only the first and fourth change what the cells do. The second and third change what you collect. A paper that conflates them will report a fivefold improvement that is partly bookkeeping, which is one of the questions to ask of the paper you read on day 5.

The menu, chosen by the product

Method	Suits a product that is	Main costs and failure modes
Organic overlay (liquid–liquid)	hydrophobic, uncharged, with a high K into a tolerable solvent	solvent toxicity and loss, emulsions, oxygen transfer through the layer, recovery by distillation
Polymer beads or resin (solid–liquid)	moderately hydrophobic; also where a solvent would be toxic or eaten	capacity ceiling, intraparticle diffusion, fouling, cost per cycle, abrasion
Gas stripping	volatile, with a reasonable Henry coefficient	huge gas flows, foaming, water and substrate co-stripping, condensation duty
Pervaporation	volatile, where gas flow is unacceptable	membrane area, fouling, modest fluxes, capital
External column in a loop	anything the cells must not contact the capture phase for	pumping, cells in the loop, shear, control of two coupled systems
Crystallisation / precipitation	sparingly soluble at some pH or temperature	nucleation control, cells in the solids, filtration
Ion exchange / electrodialysis	charged at fermentation pH — acids and amino acids	competing ions in the medium, regeneration chemicals, salt disposal

The single most useful line in that table is the last one. If your product is charged at the fermentation pH — an organic acid, an amino acid, a phosphorylated compound — a solvent is almost always the wrong answer, because the species in the broth does not partition into an organic phase at all. People try it anyway, because overlays are what the ISPR literature is mostly about, and then report a disappointing partition coefficient. Check the pK_a against the fermentation pH first; it takes one line, and it eliminates the largest category of wasted effort in this subject.

The second most useful line is the beads one. A solid phase cannot be metabolised, cannot be toxic in the way a solvent is, and can be used with mixed or undefined cultures where a solvent cannot. That is why the polymer-bead literature exists and why it is worth more of your attention than its citation count suggests.

Isotherms, and the sorbent mass you actually need

Two isotherms cover nearly everything. The linear one, $q = K_p \cdot C$, with loading q in grams per kilogram of sorbent and C the aqueous concentration, is the low-concentration limit and is adequate when you operate far below saturation. The Langmuir form,

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q = q_max · C / (K_d + C)
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adds the thing the linear form hides: a capacity ceiling. q_{max} is the loading at saturation, and K_d is the aqueous concentration at half of it — which, usefully, is the same algebra as Michaelis–Menten with the same two readings. Below K_d the sorbent behaves linearly with $K_p \approx q_{max}/K_d$; above it, adding sorbent is the only way to take up more.

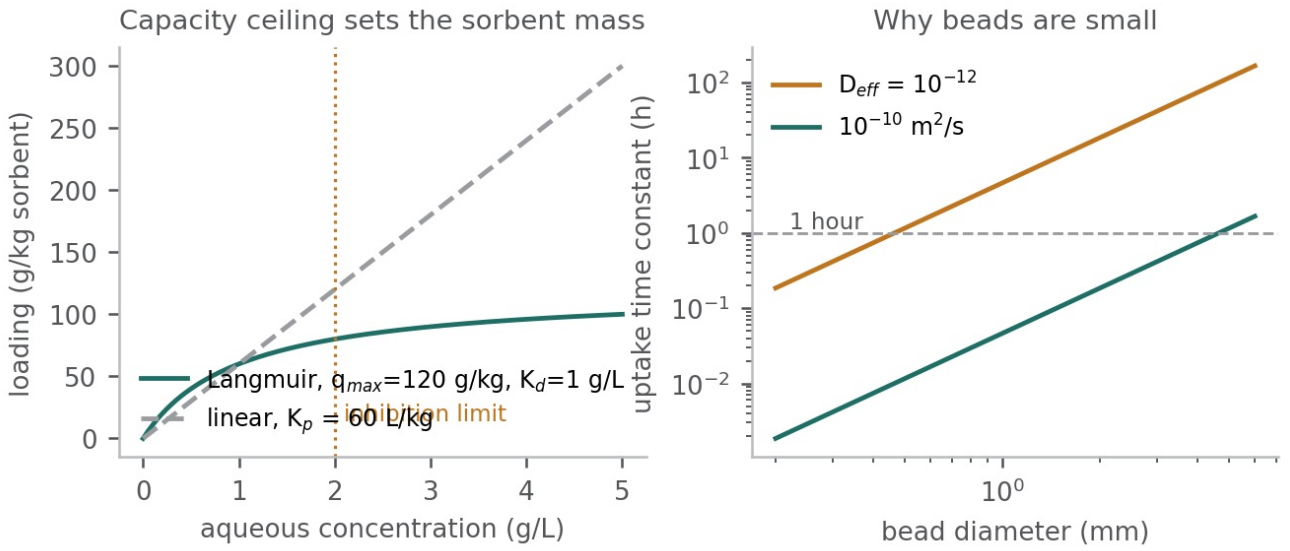


Figure 46.1 Left: a Langmuir isotherm with q_{max} = 120 g/kg and K_d = 1 g/L against its low-concentration linear approximation, with the aqueous inhibition limit marked; the sorbent must do its work to the left of that line. Right: the intraparticle uptake time constant $R^2/15D_{eff}$ against bead diameter for two effective diffusivities spanning what polymers give. Both computed for this module; the isotherm parameters are illustrative and the diffusivities are order-of-magnitude class values.

The sizing calculation is then three lines, and it is the one to do before ordering anything. You must keep the aqueous concentration at or below the inhibition limit — say 2 g/L — so the loading you can rely on is the isotherm evaluated there: with the figure's parameters, q = 120 × 2/(1 + 2) = 80 g/kg. To hold one kilogram of product you therefore need 1,000/80 = 12.5 kg of sorbent. For a 100 m³ fermentation making 20 g/L — two tonnes of product — that is 25 tonnes of resin, which is where the arithmetic usually stops a project unless the resin is cheap and reusable.

Hence the cycle question, which is the same as the solvent arithmetic in chapter 45. A resin at $50/kg used for one batch adds $625 per kilogram of product and is absurd for anything but a pharmaceutical; the same resin surviving 100 cycles adds $6.25, which is tolerable for a specialty. So the number to establish experimentally is not only the isotherm but the loading retained after tens of cycles of adsorption, elution and regeneration in real broth. Fouling by medium components, cell debris and antifoam reduces capacity cycle by cycle, and a resin that loses 2% of its capacity per cycle is finished in 35 cycles whatever its catalogue specification says.

Inside the bead: equilibrium limit against rate limit

Everything above is equilibrium. Whether equilibrium is reached is a transport question, and for a solid sorbent the resistance is almost always inside the polymer rather than in the liquid film outside it, because the solute's effective diffusivity in a dense polymer is orders of magnitude below its value in water. The characteristic time for uptake into a sphere is

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τ ≈ R² / (15·D_eff)
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and the figure plots it. For a 1 mm bead, τ is about 3 minutes at $D_{eff} = 10^{-10}$ m²/s and about 4.6 hours at 10^{-12}; for a 3 mm bead the same two diffusivities give 25 minutes and 42 hours. Since the production timescale is tens of hours, a 1 mm bead is at equilibrium with the broth throughout and the equilibrium arithmetic above holds, while a 3 mm bead in a slow polymer is not — its interior is still filling when you harvest, so you have paid for capacity you never used.

This is why beads are small, and it is also the stress test worth running on any ISPR design: double the production rate and ask whether the sorbent can still keep up. Capacity is a stock and uptake is a flow; a design can be sized correctly on the stock and fail on the flow. The rough criterion is to compare the volumetric production rate with the sorbent's uptake rate per unit volume of broth, which for a linear isotherm is roughly (phase fraction)·(K_p/τ)·(aqueous concentration). If production exceeds that, the aqueous concentration rises above equilibrium and the cells see more than you designed for — the same phenomenon that chapter 47 models properly for a liquid phase.

The polymer choice itself is a screening problem rather than a design one. Partition coefficients for one compound differ by orders of magnitude between commercial polymers, which is the central finding of the solid–liquid two-phase literature, and the differences are not reliably predictable from the product's structure. So the method is to screen a panel of a dozen polymers at a few grams each, measure the isotherm of the two or three best in real medium rather than buffer, and only then size the reactor.

Recovery, reuse, and the external loop

The product has to come back off. From a solvent, that is usually distillation or evaporation, and the governing question is whether the product is more or less volatile than the solvent and whether it survives the temperature. From a resin, it is elution with a solvent or a pH or temperature swing, and the governing question is the volume of eluate: a resin that needs ten bed volumes of methanol to strip has moved your separation problem rather than solved it. From a gas stream, it is condensation or an absorber, and the duty is set by the gas flow, which is why stripping is cheap in principle and expensive in practice.

Internal versus external is the last structural choice. An internal phase — solvent or beads in the vessel — is simple, has no pumping, and exposes the cells to the capture phase, whose abrasion can shear them and whose presence complicates oxygen transfer and foam. An external column in a recirculation loop keeps the capture phase away from the cells, can be regenerated without stopping the fermentation, and allows a toxic solvent to be used at all; the costs are the pump, the cells that spend time in the loop under different oxygen and shear conditions, the dead volume, and the fact that you are now controlling two coupled systems. For a toxic solvent or an expensive resin the loop usually wins; for a benign overlay it rarely does.

Linear isotherm	`q = K_p·C, with K_p ≈ q_max/K_d at low C`
Langmuir	`q = q_max·C/(K_d + C); the ceiling is what sizes the sorbent`
Sorbent mass	`kg sorbent per kg product = 1000 / q(C_inhibition)`
Cost per kg	`resin price / (kg product per cycle × cycles survived)`
Uptake time	`τ ≈ R²/(15·D_eff): 3 min to 4.6 h for a 1 mm bead`
Charged product	`pKa versus fermentation pH decides whether a solvent can work at all`

A paper reports that adding 10% w/v of a polymer raised the titer of a phenolic from 0.9 to 4.1 g/L, a 4.6-fold improvement. What do you check before believing it is a 4.6-fold improvement in production?

First, what the reported titer is per litre of. If the 4.1 g/L counts the product extracted from the polymer and is expressed per litre of aqueous phase while the 0.9 g/L is per litre of broth, part of the gain is bookkeeping. Second, the control: is it the same strain in the same vessel without polymer, run for the same time, or is it a historical number from a different run? Third, whether the product was previously being lost or degraded — if the phenolic has a few hours' half-life in medium, part of the gain is recovery rather than production, which is still useful but is a different claim. Fourth, substrate: if the no-polymer run left substrate unused and the polymer run consumed it all, production genuinely increased and the yield comparison will show it.

Then the scale-up questions the paper almost certainly does not answer: oxygen transfer with 10% solids in the vessel, bead abrasion and fines over a long run, fouling in real medium, how the product comes off the polymer and at what eluate volume, and the polymer cost per kilogram of product at a realistic number of cycles. None of these invalidates the result; all of them decide whether it matters.

Three things people get wrong

- ‘ISPR means a solvent.’ It means removing the product as it is made; beads, stripping, crystallisation and ion exchange are all ISPR, and for a charged product a solvent is the one that cannot work.
- ‘The isotherm is linear.’ It is, until you reach the capacity ceiling — which is exactly where a correctly sized system operates.
- ‘Capacity is what matters.’ Capacity is a stock and uptake is a flow. A big slow bead fails at the flow while looking fine on the stock.

Self-check — answer on paper first

1. Your product inhibits at 3 g/L aqueous and your sorbent gives 50 g/kg at that concentration. How much sorbent does a 10 m³ fermentation need to reach 30 g/L of whole broth, and what does that tell you?

Thirty grams per litre of 10 m³ is 300 kg of product; at 50 g/kg that is 6 tonnes of sorbent, which in a 10 m³ vessel is physically impossible — the solids would occupy more than the working volume. The calculation tells you the design is

wrong at the concept stage: either the sorbent must have a far higher loading at 3 g/L, or the removal must be external with continuous regeneration so that the same sorbent is reused many times within one batch, or the target titer must come down. This is the normal outcome of the sizing calculation and it is why it is worth doing first.

2. Distinguish an equilibrium limit from a rate limit in a solid-phase ISPR, and give the experiment that tells you which you have.

An equilibrium limit means the sorbent is saturated at the aqueous concentration you are holding — more sorbent would help, smaller beads would not. A rate limit means the sorbent is far from saturated but cannot take product up fast enough, so the aqueous concentration rises above the equilibrium value — smaller beads or more sorbent surface would help, more capacity would not. The experiment: harvest beads during the run and measure their loading against the isotherm prediction for the aqueous concentration at that moment. Loading at the predicted value means equilibrium; loading well below it means the rate limits.

3. Why can a solid phase be used with a mixed culture where a solvent usually cannot?

Because a solvent is itself an organic compound available to the organisms: in a mixed or undefined culture something will metabolise it, which consumes the capture phase, generates biomass and by-products, and changes the partitioning as the solvent composition drifts. A crosslinked polymer is not bioavailable, so it cannot be consumed, and it also cannot dissolve into the aqueous phase to reach the membrane. The trade is lower capacity per unit volume and intraparticle diffusion instead of the fast equilibration of a dispersed liquid.

What to open a book for. Again, no book. The Daugulis group's polymer papers — Prpich and Daugulis on phenol delivery (Biotechnol. Prog. 2004) and on 3-methylcatechol (Biotechnol. Bioeng. 2007), and Rehmann, Sun and Daugulis on polymer selection — are the primary literature and are short. For the transport side, any mass-transfer text's treatment of adsorption in porous media covers intraparticle diffusion better than any biology source will.

Modelling a two-phase fermentation, and fitting it to data

This is the week where your own discipline does the work. The model is four ordinary differential equations, the errors are all in the volume bookkeeping, and the honest conclusion is usually that the data identify fewer parameters than the paper claims.

The single-phase model, and the three rate laws in it

Start with one phase. Three constitutive relations and three balances:

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μ   = μ_max · S/(K_S + S) · (1 − P_a/P*)        growth: Monod, with Levenspiel product inhibition
q_P = α·μ + β·(1 − P_a/P*)·S/(K_S + S)          production: Luedeking–Piret, growth- and non-growth-associated
dX/dt  =  μ·X
dS/dt  = −(μ/Y_XS + q_P/Y_PS)·X
dP_a/dt =  q_P·X
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Each parameter does one thing, and you should be able to say what. μ_{max} and K_S are the growth kinetics; P* is the aqueous concentration at which growth stops, so it is the inhibition ceiling; α is the part of production that happens only while growing and β the part that continues in stationary phase; Y_{XS} and Y_{PS} partition substrate between biomass and product. The Levenspiel factor is the simplest inhibition form that goes to zero at a finite concentration rather than asymptotically, which is what toxicity actually looks like.

Run it with the parameters in the listing below — μ_{max} 0.40 h^{-1}, P* 2 g/L, 120 g/L substrate — and the answer is the characteristic shape of a toxicity-limited process: the aqueous product rises to 2.0 g/L, growth and production stop, and **91 of the 120 g/L of substrate is left in the broth**. The strain is not the problem; the ceiling is.

Before interpreting anything, check the substrate balance closes: substrate consumed should equal biomass formed over Y_{XS} plus product formed over Y_{PS}. In the run above it closes to better than 0.01%, which is what you want to see. That check catches sign errors and missing terms and proves nothing about whether the model is right — it is an arithmetic check, not a validation, and the distinction is worth stating in your own methods sections.

Adding the second phase, and the bookkeeping where the errors live

The second phase holds product and nothing else: no cells, no substrate. Transfer between the phases is driven by the distance from equilibrium, not by the concentration itself —

```
N = k_L a · (P_a − P_o/K)        mass transfer, per litre of aqueous phase per hour
```

— and the two new balances are

```
dP_a/dt = q_P·X − N
dP_o/dt = N · (1 − φ)/φ
```

That (1 − φ)/φ factor is the whole difficulty. N was defined per litre of aqueous phase; the organic phase is smaller or larger than the aqueous phase by exactly that ratio, so the same quantity of product arriving in the organic phase changes its concentration by a different amount. Get the factor upside down and the model runs, conserves nothing, and gives plausible-looking curves. Hence the rule: write on the page what each concentration is per litre of, and check total product — $P_a(1 − φ) + P_o φ$ per litre of whole broth — against cumulative production.

```
import numpy as np
from scipy.integrate import solve_ivp

P = dict(mumax=0.40, Ks=0.5, Yxs=0.40, Yps=0.30, alpha=0.10, beta=0.05,
         Pstar=2.0, S0=120.0, X0=0.10, K=100.0, phi=0.20, kla=50.0)

def rhs(t, y, p):
    X, S, Pa, Po = y
    S   = max(S, 0.0)
    inh = max(0.0, 1.0 - Pa/p['Pstar'])
    sat = S/(p['Ks'] + S)
    mu  = p['mumax']*sat*inh
    qp  = p['alpha']*mu + p['beta']*sat*inh
    N   = p['kla']*(Pa - Po/p['K']) if p['phi'] > 0 else 0.0
    return [mu*X,
            -(mu/p['Yxs'] + qp/p['Yps'])*X,
            qp*X - N,
            N*(1 - p['phi'])/p['phi'] if p['phi'] > 0 else 0.0]

s = solve_ivp(rhs, [0, 80], [P['X0'], P['S0'], 0.0, 0.0], args=(P,),
              method='LSODA', rtol=1e-9, atol=1e-11, dense_output=True)
X, S, Pa, Po = s.y[:, -1]
print(Pa, Po, Pa*(1-P['phi']) + Po*P['phi'])      # 0.32  32.0  6.65 g per L of broth
```

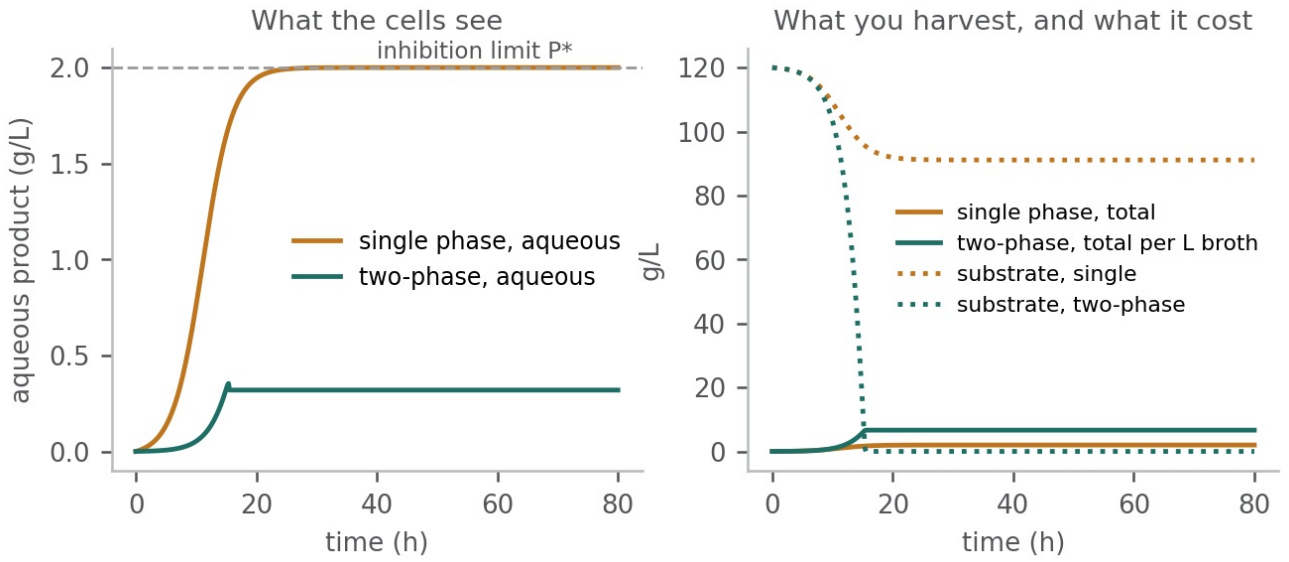


Figure 47.1 The same strain with and without a second phase (K = 100, φ = 0.2, k_La = 50 h^{-1}). Left: what the cells see — the single-phase culture sits against its inhibition ceiling, the two-phase culture never approaches it. Right: what you harvest, and the substrate it cost: the single-phase run leaves 91 g/L of substrate unused, the two-phase run consumes all of it. Computed from the model listed above; all parameters are assumed.

What the model says, and the row where more solvent is worse

With the second phase, the aqueous concentration peaks at 0.32 g/L instead of 2.0, the culture consumes all 120 g/L of substrate, and the harvest is 6.65 g per litre of whole broth against 2.0 — a 3.3-fold gain, obtained without touching the strain. That is the case for ISPR in one sentence, and it is also a warning: the gain came from removing a ceiling, so it is bounded by what the strain and the substrate can actually make, not by the partition coefficient.

Sweep the solvent fraction and the point becomes sharp:

φ	R = 1 + Kφ/(1−φ)	peak aqueous (g/L)	harvest per L of broth (g/L)
0.05	6.3	1.33	7.89
0.10	12.1	0.69	7.48
0.20	26.0	0.32	6.65
0.30	43.9	0.19	5.82
0.50	101.0	0.08	4.15

R rises twentyfold across that range and the harvest *falls* by half. Once the second phase is large enough to keep the aqueous concentration well below P*, extra solvent buys nothing — the ceiling is no longer binding — and every additional litre of it displaces a litre of aqueous phase containing the cells that make the product. The optimum here is

around 5%, far below what the capture arithmetic alone would suggest. This is the row the tracker asks you to explain, and the explanation is that two different quantities are being optimised: R maximises capture per litre of aqueous phase, while a plant maximises product per litre of vessel.

The equilibrium shortcut, and when it fails

If transfer is fast relative to production, the phases stay at equilibrium, $P_o = K \cdot P_a$, and the whole second phase collapses into a single algebraic factor: the inhibition ceiling behaves as though it were $P^* \cdot R$ with $R = 1 + K\varphi/(1 - \varphi)$. You do not need the fourth differential equation at all. Most of the time this is the right model, and deriving it is worth doing once so that you trust it.

When does it fail? Define the relaxation rate at which an offset from equilibrium decays:

```
λ = k_L a · (1 + (1 − φ)/(K·φ))
```

A production rate $q_P X$ must be carried across the interface, and carrying it requires an offset of roughly $q_P X/\lambda$. So the shortcut is good when that offset is small compared with P^*, and bad when it is not. The model confirms the estimate: the measured offset tracks $q_P X/\lambda$ across three orders of magnitude of $k_L a$.

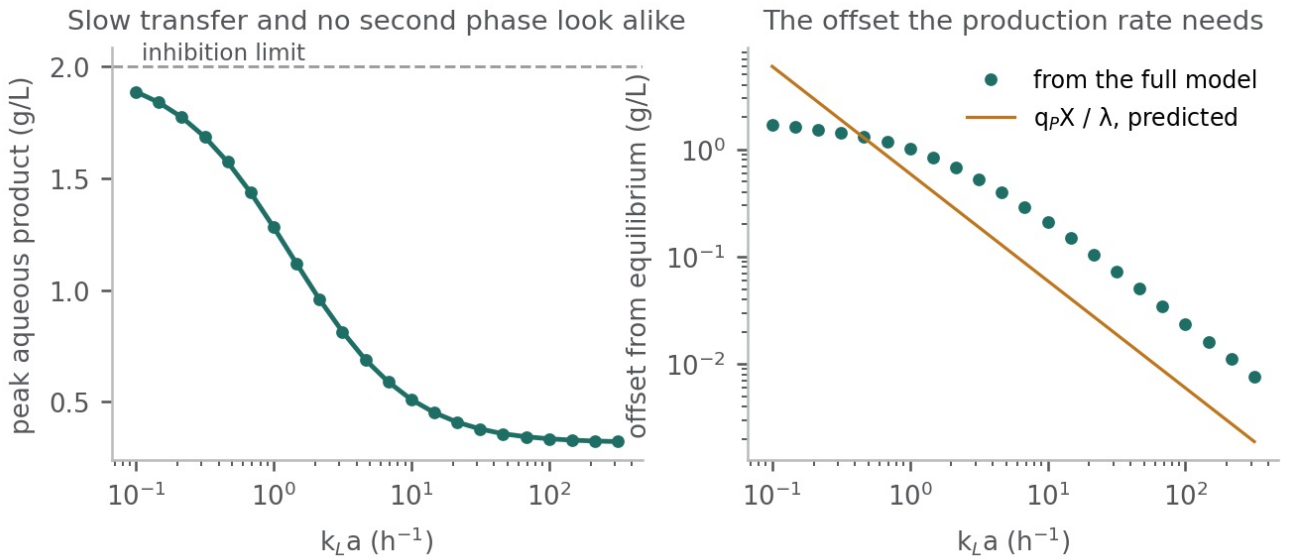


Figure 47.2 Left: the peak aqueous concentration the cells experience against the transfer coefficient. At $k_La = 0.1\ h^{-1}$ the culture reaches 1.89 g/L — almost the inhibition limit — despite a second phase with a capture factor of 26; at $10\ h^{-1}$ it reaches 0.51 and at $100\ h^{-1}$, 0.34. Right: the offset from equilibrium against the prediction q_PX/λ. Computed from the model; q_PX is taken at its mid-run value of about 0.62 g/L/h.

The practical reading is the left panel, and it is the result of the week. **Slow transfer and no second phase look the same to the cells.** A process with an excellent partition coefficient and a poorly dispersed solvent — a thick unmixed overlay, large droplets, or beads too big — delivers the toxicity of a single-phase process while incurring all the cost of the second one. Dispersion is therefore not a detail of implementation, it is part of the design, and the measurement that settles it is the aqueous concentration during the run, not the total at harvest.

A numerical note for week 12's benefit: a large k_La makes this system stiff, because transfer then equilibrates in minutes while the fermentation runs for days. That is exactly the structure chapter 12 described, and it is why every listing here uses LSODA. If you must explore k_La over orders of magnitude, use the equilibrium shortcut in the fast regime instead of integrating through it — which is model reduction doing its job.

Fitting, and what the data actually identify

Now the uncomfortable part. Generate synthetic aqueous-phase data from the two-phase model, add 2% noise, and fit K and k_La back. The fit is excellent to the eye and returns $K = 107 \pm 2$ — within 7% of the true 100 — and $k_La = 28 \pm 11$ against a true value of 50. Profile the residual against k_La and the reason is obvious: it falls steeply up to about $10\ h^{-1}$ and is then essentially flat, so any value above 15 fits the data equally well.

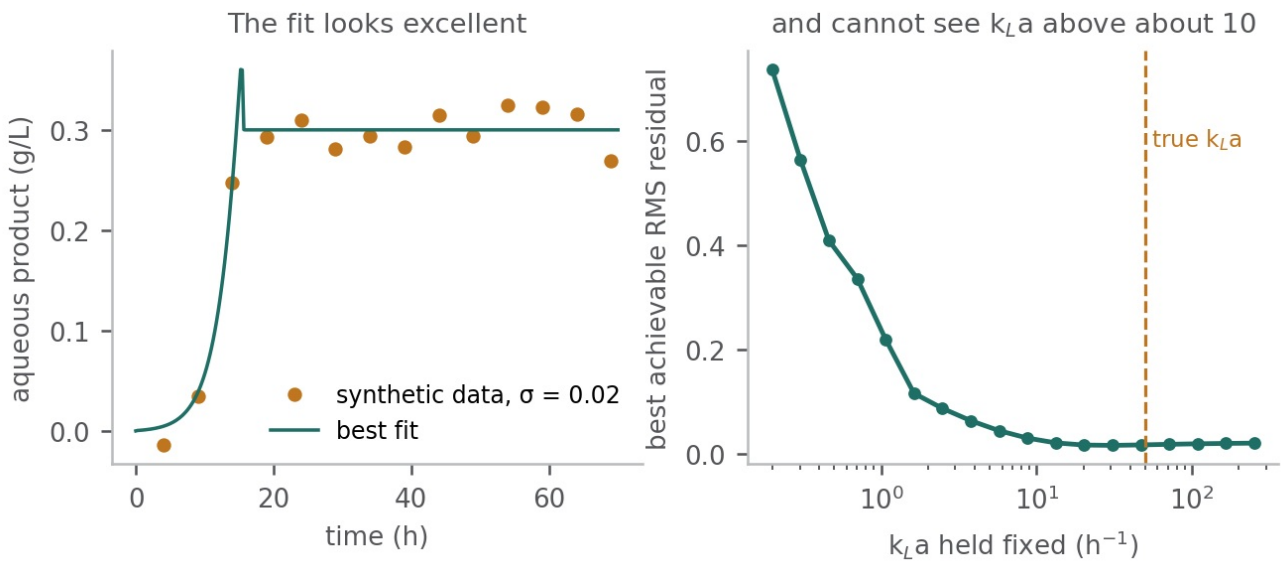


Figure 47.3 Left: the fit to synthetic aqueous-phase data, which looks excellent. Right: the profile of the best achievable residual against k_La held fixed, showing that the data constrain it only up to about 10 h^{-1}. The partition coefficient is determined to a few per cent; the transfer coefficient is not determined at all in the fast regime. Computed for this module with `least_squares`.

That is not a defect of the optimiser, it is the physics: once transfer is fast enough to hold equilibrium, the system's behaviour depends on K and φ and not on how fast equilibrium is reached, so the data contain no information about k_La. The methods-section sentence that follows is the deliverable of this week: *K was estimated from aqueous-phase concentrations by least squares, with the interval from the covariance of the fit; k_La was not identifiable above approximately 10 h^{-1} and was therefore fixed at a value in the fast-transfer regime, which the data cannot distinguish from any larger value.* That sentence is more credible than a table of four parameters with tight intervals, and reviewers who know the subject will read it as a sign you understood your own model.

Two further habits. Fit from two different starting points: agreement is weak evidence of a unique optimum, disagreement is strong evidence of a problem. And ask what a second experiment would add — here, a run at low k_La, where the offset is large and the parameter is visible, which is the experiment the sensitivity analysis of chapter 16 would have told you to do.

Growth	`μ = μ_max·S/(K_S+S)·(1 − P_a/P*)`
Production	`q_P = α·μ + β (Luedeking–Piret)`
Transfer	`N = k_L a·(P_a − P_o/K); the driving force is distance from equilibrium`
Bookkeeping	`dP_o/dt = N·(1 − φ)/φ; say per litre of what, every time`
Harvest	`per litre of broth = P_a(1 − φ) + P_o·φ`
Shortcut	`fast transfer → ceiling × R, R = 1 + Kφ/(1 − φ)`
When it fails	`offset ≈ q_P X/λ, λ = k_L a·(1 + (1 − φ)/(Kφ))`

A collaborator's two-phase model predicts a 26-fold titer gain from an overlay at φ = 0.2, K = 100. Your experiment gives 3.2-fold. Reconcile them.

They are both right, and the factor of eight between them is three separate things. First, the 26 is R — the multiplication of the *aqueous ceiling*, per litre of aqueous phase — and the experiment measured product per litre of broth; that accounts for a factor of (1 − φ) = 0.8 on its own. Second, R only pays off if the ceiling is what binds all the way to the end. In my model the second phase removes the ceiling after a few hours and the process becomes substrate-limited instead: the gain is then set by how much product the strain can make from the available substrate, which was 6.65 against 2.0 g/L — a factor of 3.3, almost exactly what your experiment found. Third, if the actual k_La or dispersion is poor, the cells still see a significant aqueous concentration and part of the ceiling remains in force.

How to tell which applies, in one measurement: the residual substrate at harvest. If the two-phase run consumed all the substrate, you are substrate-limited and the gain cannot be increased by more solvent — the next intervention is the strain or the feed. If substrate remains and the aqueous concentration was high during the run, transfer or dispersion is the limit. If substrate remains and the aqueous concentration stayed low, something else stopped the culture and neither model applies.

Three things people get wrong

- ‘More solvent is better.’ Past the point where the ceiling stops binding, extra solvent only displaces productive volume. In the model the optimum was 5%.
- ‘The transfer term is $k_La \cdot P_a$.’ It is $k_La \cdot (P_a - P_o/K)$. A driving force is a distance from equilibrium.
- ‘The model fits, so the parameters are measured.’ In the fast-transfer regime the data carry no information about k_La at all. Profile it and say so.

Self-check — answer on paper first

1. Write the two-phase product balances from memory and state what each concentration is per litre of.

$dP_a/dt = q_PX - k_La(P_a - P_o/K)$, with P_a and X per litre of aqueous phase; $dP_o/dt = k_La(P_a - P_o/K) \cdot (1 - \varphi)/\varphi$, with P_o per litre of organic phase. The factor converts a quantity expressed per litre of aqueous phase into a concentration change in a phase of different size. The harvest figure, per litre of whole broth, is $P_a(1 - \varphi) + P_o\varphi$.

2. Your aqueous concentration sits at 1.9 g/L during a run with a second phase whose capture factor is 26. What is happening and what do you change?

Transfer is too slow to carry the production rate, so the aqueous phase runs far from equilibrium with the organic phase and the cells experience almost the full inhibition ceiling. The offset needed is q_PX/λ with $\lambda = k_La(1 + (1 - \varphi)/(K\varphi))$, so the fix is anything that raises k_La: better dispersion through agitation, smaller droplets or beads, or more interfacial area at the same phase fraction. Adding solvent is the wrong move — it raises R, which is not what is binding.

3. Why does the substrate-balance closure check prove so little, and what would a real validation require?

Because closure is a property of the arithmetic, not of the biology: a model with the wrong rate laws, wrong parameters and wrong mechanism still conserves mass if the stoichiometric coefficients are consistent. It catches sign errors and missing terms, which is worth having. A real validation requires prediction of data the model was not fitted to — a different phase fraction, a different initial substrate, a different k_La — with the parameters fixed at the values obtained from the first dataset.

What to open a book for. Nothing covers this. Newman et al., Biotechnol. Bioeng. (2006) is the two-phase case study to read for real numbers, and the 2012 Bioprocess Biosyst. Eng. paper on substrate transport in two-phase partitioning bioreactors is the closest thing to a modelling reference. The SciPy documentation for `solve_ivp` and `least_squares` is, for this week, more useful than any textbook — which is itself the reason this chapter exists.

Chapter 48 · tracker week 48

Cell immobilisation: diffusion, dead cores and continuous operation

Immobilising cells retains biomass. It does not remove product — that was the last two chapters — and confusing the two is the commonest error in this area. The question this week answers is whether retention is worth what it costs you in oxygen.

What it is for, and what it is not for

Entrapping or attaching cells does four things, and you should be able to say which of them you are buying.

- **Retention.** Biomass is no longer washed out by the feed, so you can run continuously at a dilution rate above the organism's maximum growth rate. This is the main reason, and it is the only one that changes what is thermodynamically possible rather than merely convenient.
- **Reuse of the biocatalyst.** For a conversion rather than a growth process — a whole-cell biotransformation — the beads are a catalyst you recover and use again.
- **Easier separation.** Millimetre beads settle or are screened out; cells are centrifuged. For a clean broth this is worth real money downstream.
- **Some protection.** From shear, and partly from inhibitors, because the bead interior is a different microenvironment from the bulk — which is also the source of every problem below.

What it does not do is lower the product concentration in the broth. If your problem is product toxicity, immobilisation does not solve it and may make it worse, since the product must diffuse out through the same polymer. And it requires cells that work while largely non-growing, because growth inside a bead bursts it. So the first question about any candidate product is whether its production is growth-associated: in the Luedeking–Piret language of chapter 47, immobilisation needs a large β and tolerates a small α. A strictly growth-associated product is a poor candidate however attractive continuous operation looks.

Materials, and the failure each one has

Material	Gentleness	Strength	The specific weakness
Calcium alginate	Very gentle — beads form by dropping into $CaCl_2$ at room temperature	Weak	Dissolves when calcium is removed: phosphate buffers, citrate and other chelators, and high sodium all strip the crosslinks. Rules out most standard media without modification
κ-Carrageenan	Gentle, gels with potassium on cooling	Weak to moderate	Needs K^+ maintained; softens with temperature; similar ion sensitivity
Polyvinyl alcohol	Moderate — freeze–thaw or boric acid routes	Tough, reusable	The boric acid route is harsh for some strains; freeze–thaw costs a cycle; more opaque to oxygen transfer design
Polyacrylamide	Harsh	Tough	Acrylamide monomer is toxic to the cells being entrapped and to the operator; largely historical for living cells

The alginate–phosphate incompatibility deserves emphasis because it catches everyone once. A defined medium with phosphate as a buffer and a nitrogen or trace-element source will slowly dissolve calcium alginate beads, and the symptom is cells appearing in the broth over hours to days, which looks like leakage or bead breakage and is actually chemistry. Either reformulate the medium, add calcium continuously, or use a tougher polymer.

Diffusion inside the bead: the Thiele modulus

A bead is a catalyst pellet with live catalyst in it, so the analysis is the one you already know. For a first-order reaction in a sphere of radius R with effective diffusivity D_{eff}, a steady-state shell balance gives the concentration profile, and the dimensionless group that governs it is

```
φ = R·√(k/D_eff)            (Thiele modulus: reaction rate ÷ diffusion rate)
```

with the effectiveness factor — the rate the bead achieves divided by the rate the same cells would achieve fully exposed —

```
η = (3/φ²)·(φ·coth φ − 1),    η → 1 for φ ≪ 1,    η → 3/φ for φ ≫ 1
```

Read the two limits physically. Small φ means reaction is slow relative to diffusion: the substrate reaches the centre at nearly bulk concentration and the whole bead works, η ≈ 1. Large φ means reaction is fast relative to diffusion: the substrate is consumed in a thin outer shell, η ≈ 3/φ, and the interior is a passenger. In that regime halving the bead diameter doubles the rate, which is the design lever.

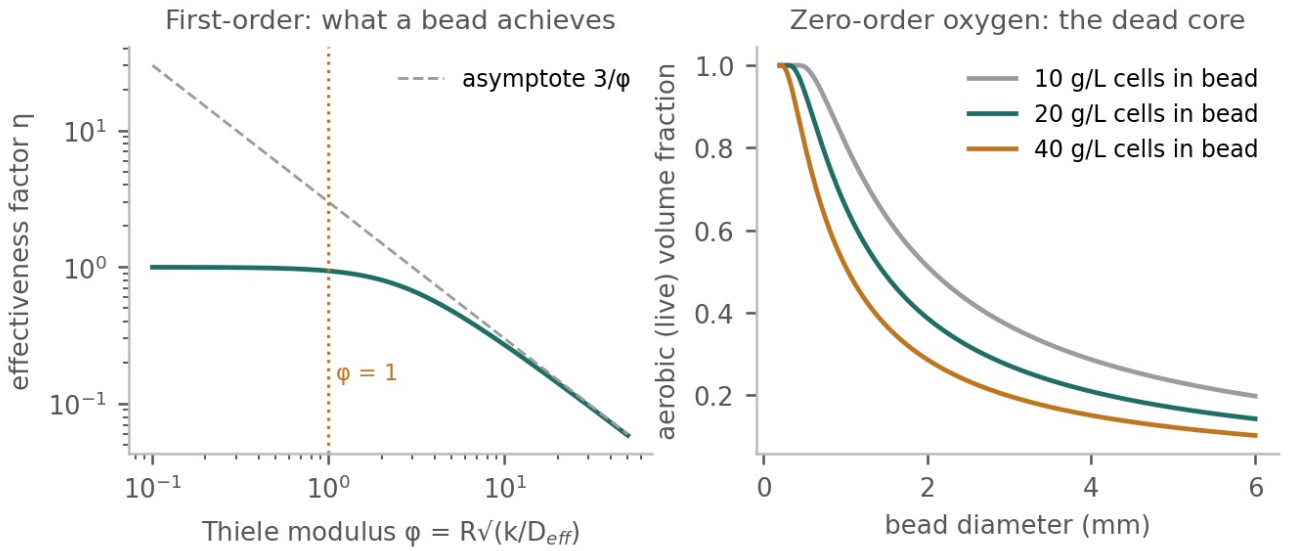


Figure 48.1 Left: the sphere effectiveness factor against the Thiele modulus with its 3/φ asymptote. Right: the aerobic fraction of a bead against diameter for three cell loadings, from the zero-order dead-core calculation with air-saturated bulk oxygen (0.21 mol/m³), $D_{eff} = 1.5 \times 10^{-9}$ m²/s and a specific uptake of 5 mmol O_2 per gram of cells per hour. Assumptions stated; arithmetic computed.

Dead cores, and why it is always oxygen

Oxygen consumption is better described as zero order than first order, because cells respire at their maximum rate down to very low oxygen concentrations and then stop. With zero-order consumption at rate k_0 per unit bead volume, the oxygen is exhausted at a finite depth rather than decaying asymptotically, and that penetration depth comes from the same shell balance:

```
L = √(2·D_eff·C_s / k₀)
```

Put numbers in. A bead loaded at 20 g/L of cells respiring at 5 mmol O_2 per gram per hour consumes 100 mol·m^{-3}·h^{-1}, or 0.028 mol·m^{-3}·s^{-1}. With air-saturated bulk oxygen at 0.21 mol/m³ and $D_{eff} = 1.5 \times 10^{-9}$ m²/s, L = **151 µm**. At 10 g/L it is 213 µm and at 40 g/L it is 106 µm. So in a 2 mm bead at 20 g/L, only the outer 0.15 mm is aerobic and the live fraction is $1 - ((R - L)/R)^3 \approx 0.39$ — most of your catalyst is sitting in an anaerobic core doing nothing, or doing something you did not design.

Now do glucose for comparison, because the contrast is the lesson. At 20 g/L in the broth, glucose is 111 mol/m³ — more than 500 times the oxygen concentration — and even with a lower diffusivity in the gel and a consumption rate of 10 mmol per gram per hour, the penetration depth comes out at about **1.1 mm**. Glucose reaches the centre of a 2 mm bead comfortably; oxygen does not come close. The reason is entirely the solubility: oxygen's driving force is small because its saturation concentration in water is small, and no amount of agitation changes that. Agitation raises the transfer rate to the bead surface, not the concentration at it.

Three levers shrink a dead core, in increasing order of cost. Smaller beads: L is fixed, so the aerobic shell is a larger fraction of a smaller sphere — going from 2 mm to 0.8 mm at 20 g/L takes the live fraction from about 0.39 to about 0.75. Lower cell loading: fewer cells consume oxygen more slowly and L rises as the inverse square root, so halving the loading raises L by 41% — but you have also halved the catalyst. More oxygen in the bulk: enriched air or elevated pressure raises C_s, and L rises as its square root, which works and costs utilities and capital.

And do not take the dead core from the model alone. The measurement is a microelectrode profile through a bead, which is standard in the biofilm literature, or a redox or oxygen-sensitive dye that reports the anaerobic zone visually. The model tells you what to look for; it does not establish that it is there, since D_{eff} in your particular gel at your particular cell loading is the parameter you are least sure of.

Continuous operation: what retention actually buys

Here is the comparison that decides the week, and it is less favourable to immobilisation than the literature's enthusiasm

suggests. Free cells in a chemostat wash out above a critical dilution rate, which for μ_{max} = 0.5 h^{-1}, K_S = 0.5 g/L and 20 g/L feed is 0.488 h^{-1}. Below it, biomass is $X = Y_{XS}(S_0 - S)$ with $S = K_S D/(\mu_{max} - D)$, and volumetric productivity is $q_P X$ with $q_P = \alpha D + \beta$. That curve peaks at **0.65 g/L/h at D ≈ 0.39 h^{-1}** and then collapses at washout.

Immobilised cells do not wash out, so productivity is set by capacity rather than by growth: beads occupying 30% of the volume at 40 g/L of cells inside, with an effectiveness factor of 0.45 from the dead-core calculation, give an effective 5.4 g/L of working biomass, and at β = 0.05 g per gram per hour that is **0.27 g/L/h — flat in D** once the feed supplies enough substrate.

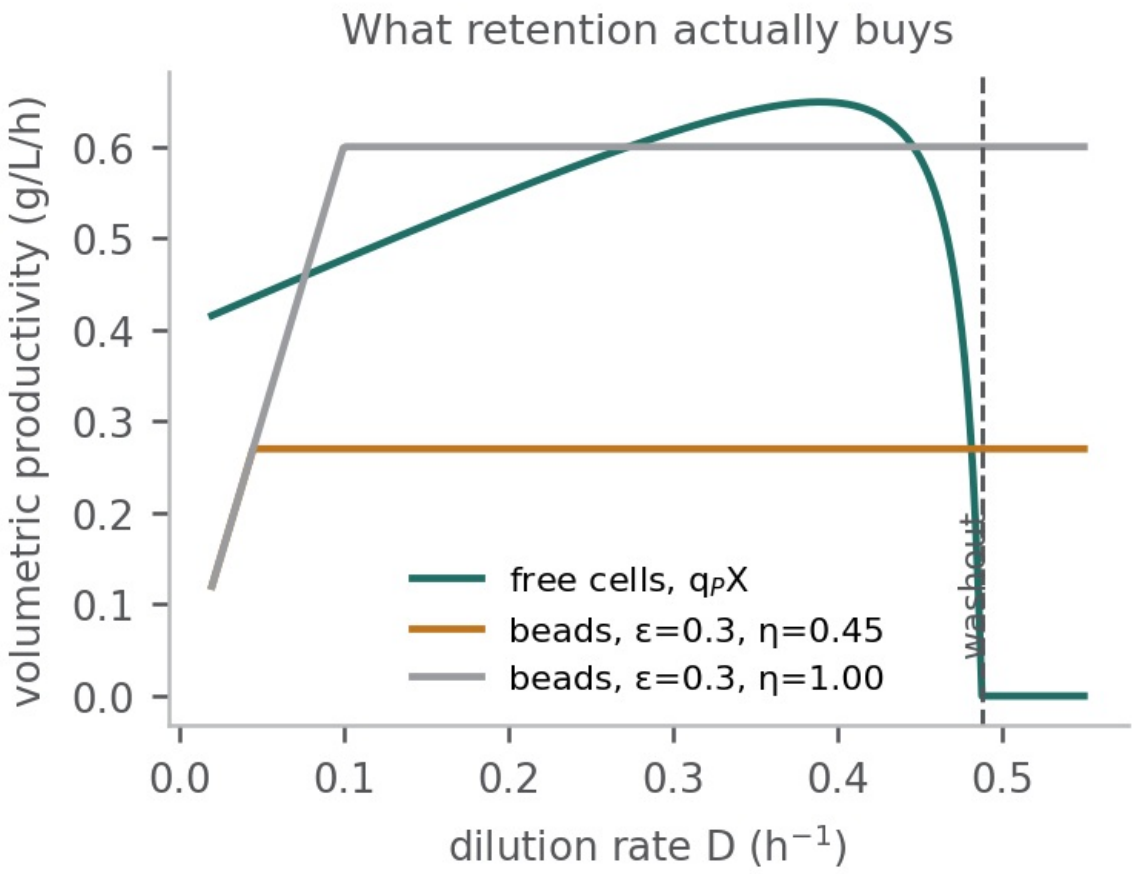


Figure 48.2 Volumetric productivity against dilution rate for free cells and for two immobilised cases, with the free-cell washout marked. The immobilised lines rise with the substrate the feed supplies and then flatten at the capacity of the retained cells. Parameters in the text; the comparison is model arithmetic and its conclusion depends entirely on β and η.

So free cells win, by more than twofold, and that is the normal result. The reason is visible in the arithmetic: retention gives you biomass that cannot wash out, but the retained cells are non-growing, so they produce only at β, and a fraction η of them are in an anaerobic core. A free-cell chemostat run just below washout has nearly all of its biomass producing at αD + β with full oxygen access.

What would have to change for immobilisation to win? Any of four things, and they are the questions to ask of any immobilisation proposal. The non-growth-associated rate β would have to be high — a product made efficiently by resting cells, which is exactly the whole-cell biotransformation case. The product would have to inhibit growth strongly, so that free cells cannot be maintained at a high dilution rate at all while resting cells keep producing. The organism would have to grow slowly, so that washout constrains free cells at an uneconomic dilution rate — immobilisation is far more attractive for a slow-growing organism than for *E. coli*. Or downstream separation of free cells would have to be unusually expensive relative to the productivity lost.

There is also a stability argument that the arithmetic above hides. The free-cell optimum sits just below washout, and that is an uncomfortable place to run a plant: a transient in the feed rate, a temperature excursion or a small drop in μ_{max} from a medium batch takes you over the cliff and the vessel empties itself over a few hours. A retained-biomass system has no cliff. Trading some productivity for an operating point that cannot catastrophically fail is a legitimate engineering decision, and it is the strongest honest argument for immobilisation in a growth-based process.

How these systems fail

- **Bead breakage.** Cells grow inside, gas bubbles form from respiration or fermentation, and the bead cracks. Monitor by sieving samples and counting intact beads, and by free cell density in the effluent.
- **Leakage.** Cells escape from the surface layer continuously, which is normal at a low rate and a failure at a high one. A rising effluent cell count that is not accompanied by broken beads is leakage rather than breakage, and the distinction matters because the fixes differ.
- **Fouling and channelling.** In a packed bed, biomass and precipitate block the interstices, flow finds a path, and part of the bed stops contributing. The signature is a rising pressure drop with a falling conversion — monitor both, and a tracer residence-time distribution will show channelling directly.
- **Lost working volume.** Beads at 30% of the vessel displace 30% of the liquid, exactly as a solvent does in chapter 45. It belongs in the productivity comparison, and in the figure above it already is.
- **Oxygen at scale.** A packed bed cannot be sparged effectively, a fluidised bed can but abrades the beads, and an airlift is gentle but has limited transfer. The reactor form is chosen by the oxygen requirement, and for a strictly aerobic process that choice is usually the binding constraint.

Thiele modulus	$\varphi = R\cdot\sqrt{(k/D_{eff})}$
Effectiveness	$\eta = (3/\varphi^2)(\varphi\cdot\coth\varphi - 1)$; $\eta \approx 3/\varphi$ for large φ
Dead core	$L = \sqrt{(2\cdot D_{eff}\cdot C_s/k_0)}$; live fraction $= 1 - ((R-L)/R)^3$
Oxygen vs glucose	151 µm against 1.1 mm, at the stated loadings — it is the solubility
Free-cell washout	$D_{crit} = \mu_{max}\cdot S_0/(K_S + S_0) = 0.488\ h^{-1}$ here
Retained productivity	$\beta\cdot X_{bead}\cdot\varepsilon\cdot\eta$, flat in D; 0.27 against 0.65 g/L/h free

You are asked to immobilise a strain making a growth-associated product in 3 mm alginate beads in a packed bed, fed a phosphate-buffered defined medium. Write the three objections, in the order you would raise them.

First, the product. Growth-associated production means α is large and β is small, and immobilised cells must be largely non-growing or they destroy the beads. The process is therefore asking resting cells to do something they do at a low rate, and the productivity comparison above will favour free cells by a wide margin. This objection is fatal to the concept rather than to the implementation, so it goes first.

Second, the beads. Three millimetres with any realistic cell loading gives an aerobic shell of about 0.15 mm, so roughly three quarters of the entrapped biomass sits in an anaerobic core. If the process is aerobic, you are paying for catalyst that is not working; if the organism is facultative, the core will ferment and excrete by-products into your product stream. Submillimetre beads, or a different reactor, would be needed.

Third, the chemistry. Calcium alginate in phosphate buffer dissolves: phosphate sequesters the calcium that crosslinks the gel. The beads will soften and shed cells over days, which in a packed bed also means fouling and channelling. Either the medium's buffer changes, calcium is fed continuously, or the material changes to PVA.

What I would propose instead, if continuous operation is the real objective: cell retention by membrane filtration or settling rather than entrapment, which keeps the biomass growing and fully oxygenated, has no diffusion limit and no dead core, and achieves the same decoupling of dilution rate from growth rate. Immobilisation is the right answer for a resting-cell biotransformation, not for a growth-associated fermentation.

Three things people get wrong

- ‘Immobilisation and ISPR are both ways of improving a toxic-product process.’ Immobilisation retains biomass; it does nothing to the product concentration the cells experience.
- ‘Glucose limitation inside the bead is the problem.’ It is oxygen, by an order of magnitude, and for the same reason in every system: solubility.
- ‘Retention beats washout, so it beats free cells.’ Retained cells are non-growing and partly anaerobic. Do the productivity comparison; it usually favours free cells.

Self-check — answer on paper first

1. Derive the oxygen penetration depth for zero-order consumption and compute it for a bead at 30 g/L of cells, using the chapter's other assumptions.

A shell balance with constant consumption gives $D_{eff}\cdot d^2C/dx^2 = k_0$ in the slab limit, and integrating with $C = C_s$ at the surface and both C and dC/dx vanishing at the front gives $L = \sqrt{(2\cdot D_{eff}\cdot C_s/k_0)}$. At 30 g/L and 5 mmol·g^{-1}·h^{-1}, $k_0 = 150$ mol·m^{-3}·h^{-1} $= 0.0417$ mol·m^{-3}·s^{-1}, so $L = \sqrt{(2 \times 1.5 \times 10^{-9} \times 0.21/0.0417)} \approx 123$ µm. In a 2 mm bead that is a live fraction of $1 - (0.877)^3 \approx 0.33$.

2. A colleague proposes doubling the cell loading in the beads to double the productivity. What happens?

Productivity rises by much less than twofold and may not rise at all. Doubling k_0 reduces the penetration depth by a factor of √2, so the aerobic shell thins; the live biomass per bead is roughly the loading times the live fraction, and in the large-bead limit — where the live volume is a thin shell of thickness L — the product of loading and L scales as √(loading), so the gain is about 41% at best. It also raises the oxygen demand per bead, so the bulk oxygen concentration falls and the whole calculation shifts further against you. Smaller beads are the better lever.

3. State the honest condition under which immobilisation beats a free-cell chemostat.

When the non-growth-associated productivity of the retained cells, multiplied by the retained biomass and by the effectiveness factor, exceeds the free-cell volumetric productivity at its best dilution rate — which requires a product made well by resting cells, or an organism whose growth rate is too low to permit an economic dilution rate, or a product that inhibits growth so strongly that free cells cannot be maintained at all. Add to that the stability argument: a retained system has no washout cliff, which can justify accepting lower productivity for an operating point that cannot fail catastrophically.

What to open a book for. Here a book does exist, and it is a chemical engineering one: any reaction-engineering text's treatment of the Thiele modulus and effectiveness factor for catalyst pellets covers the mathematics of this week exactly, and better than the biological literature does. For the biology, the reviews of alginate and PVA immobilisation and of oxygen limitation in immobilised-cell beads are the sources; read them for the materials and the failure modes rather than the theory.

Closing notes to part 3

The four weeks in one page

Week 45. Three different limits — toxicity, export, loss — look identical on a plot and need three different interventions. Three experiments separate them, and none takes a week. A plateau at q/k is a loss, not a stopped pathway.

Week 46. The product's properties choose the removal method, and the first filter is charge: if it is ionised at fermentation pH a solvent cannot work. For a sorbent, the Langmuir ceiling at the inhibition concentration sizes the mass, the cycle count sets the cost, and intraparticle diffusion decides whether the capacity you bought is reachable.

Week 47. Four ODEs, one transfer term driven by the distance from equilibrium, and a volume factor that is where the errors live. Past the point where the ceiling stops binding, more solvent lowers output per litre of vessel. Slow transfer and no second phase look the same to the cells. And in the fast-transfer regime the data do not identify k_La at all, which you should say rather than hide.

Week 48. Immobilisation retains biomass and does nothing about product concentration. Oxygen penetrates about 150 µm at realistic loadings while glucose penetrates a millimetre, so most of a large bead is anaerobic. The honest productivity comparison usually favours free cells, and the real argument for retention is the absence of a washout cliff.

On the numbers in this part

Every figure here was computed by a script and every parameter is stated. The two-phase model of chapter 47 is listed in full and generated figures 47.1, 47.2 and 47.3 as well as the phase-fraction table and the identifiability result; its parameters are assumed, not measured, and the captions say so. The transport numbers in chapters 46 and 48 use order-of-magnitude effective diffusivities — the quantity you are least sure of in any real system — and the conclusions that depend on them are stated as comparisons (oxygen against glucose, 1 mm against 3 mm beads) rather than as absolute predictions, because comparisons survive an error in D_{eff} that absolute numbers do not. The efflux-pump result is from the published screen and is quoted as a base rate rather than a prediction.

Where a number here would decide a purchase or an experiment, measure it: the partition coefficient and the isotherm in real medium rather than buffer, the loss constant by a hold experiment with no cells, and the effective diffusivity by the uptake curve of a single bead. Each is a day's work and each replaces an order-of-magnitude assumption with a number.

What is still to come

Working forward in tracker order from here, the remaining weeks with no textbook behind them are techno-economic analysis and life-cycle assessment (week 49), analytical method validation (50), the product-class weeks (53 to 57, including the PHA case study and the C1 and CO_2 feedstocks), and then the host weeks — yeasts in 58, the Gram-positives *C. glutamicum* and *B. subtilis* in 59, and the phototrophs in 60, with the panorama and cell-free systems in 61. Those host chapters are the ones that will need the most original work, because what you need from them is not a species description but a comparison: what each host costs you in tools, growth rate, cofactor supply, secretion and regulatory acceptance, against the arithmetic of parts 2 and 3. That is the next part, and the immobilisation and ISPR results here are inputs to it — week 58's compartment arithmetic and week 60's photobioreactor light penetration are the same transport calculations as chapter 48's dead core, with light in place of oxygen.